

Formation Engine Methodology

The Fuzzy Ball Model

Multi-Layer Directional Descriptor Visualization
of Coarse-Grained Atomistic Beads

Spherical Harmonic Surface Fields, Anisotropic Shell Rendering, and
Environment-Responsive Visual Inspection

VSEPR-SIM Project

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Abstract

This document describes the *Fuzzy Ball Model*, a multi-layer visual representation framework for coarse-grained (CG) beads in the Formation Engine. Rather than treating each CG bead as a featureless point mass, the fuzzy ball model preserves and exposes the directional, anisotropic, and environment-responsive character of the underlying atomistic fragment. Each bead is rendered as a translucent spherical shell whose effective radius encodes atomistic extent, whose surface carries a three-channel descriptor field expanded in real spherical harmonics, and whose interior optionally reveals the source atomistic skeleton. Eight composable view modes—*Shell*, *Scaffold*, *SurfaceState*, *InternalRef*, *Skeleton*, *Cutaway*, *Residual*, and *Comparison*—permit progressive disclosure from macroscopic overview to fragment-level inspection. A five-level scale hierarchy (*SceneCloud* $\rightarrow$ *FragmentCompare*) governs level-of-detail transitions. The model is deterministic, fully inspectable, and exports structured data suitable for scientific reporting. Implementation: `coarse_grain/vis/bead_visual_record.hpp` and `coarse_grain/vis/bead_inspector_view.hpp`.

Contents

§1 Introduction

Coarse-grained molecular simulation replaces groups of atoms with single interaction sites (beads), reducing degrees of freedom by one to two orders of magnitude. The standard visual representation of a CG bead is a uniform sphere, which discards all information about internal geometry, anisotropy, and interaction directionality.

The **Fuzzy Ball Model** restores this information by treating each bead not as a hard marble but as a *semi-transparent shell with directional structure*—a “fuzzy ball.” The model achieves this through three complementary data layers:

1. **Layer 1 — Effective Shell.** A spherical shell whose radius, opacity, colour, and deformation encode the bead’s effective extent, mapping confidence, chemical class, and principal anisotropy.
2. **Layer 2 — Surface-State Field.** A structured (θ, φ) grid of steric, electrostatic, and dispersion descriptor values painted as a spherical heatmap over the shell.
3. **Layer 3 — Internal Structural Reference.** The original atomistic fragment (atoms, bonds, orientation markers) rendered inside the translucent shell for direct comparison.

These layers are combined through eight selectable *view modes* that control which layers are active and at what opacity. A five-level *scale hierarchy* determines how much detail is rendered at each zoom level, from a distant scene cloud to a close-up fragment comparison.

The model is implemented entirely within the `coarse_grain/vis/` subsystem and consumes the `BeadVisualRecord` adapter object produced by `build_visual_record()`.

§2 Theoretical Foundation

2.1 Effective Radius and Anisotropy

Each CG bead maps a molecular fragment to a single centre $\mathbf{r}_B$. The *effective radius* R_{eff} quantifies the spatial extent of this fragment. When a surface descriptor probe radius is available,

$$R_{\text{eff}} = r_{\text{probe}}, \quad r_{\text{probe}} > 0, \quad (1)$$

otherwise a mass-based fallback is used:

$$R_{\text{eff}} = 1.5 \left(\frac{M_B}{12} \right)^{1/3}, \quad R_{\text{eff}} \in [0.5, 4.0] \text{ \AA}. \quad (2)$$

The bead is not spherically symmetric in general. The inertia tensor of the source fragment yields three principal eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$. Their ratios define an *anisotropy vector*,

$$\mathbf{a}_B = (\lambda_1, \lambda_2, \lambda_3), \quad (3)$$

which deforms the shell rendering from a perfect sphere toward an ellipsoid.

2.2 Local Frame Assignment

A local coordinate frame $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3\}$ is assigned to each bead via one of several strategies, selected automatically based on fragment topology:

Table 1: Frame-method selection rules.

Method	Condition	Description
InertiaFit	Full surface data available	PCA of inertia tensor
RingPlaneFit	Fragment is cyclic / aromatic	$\hat{\mathbf{e}}_3$ = ring-plane normal
MetalCoordination	Fragment is organometallic	Principal axis of coordination shell
PCA_Fallback	Orientation normal only	Gram–Schmidt from $\hat{\mathbf{n}}$
SingleAtom	Fragment has ≤ 1 atom	Degenerate; confidence = 0

Frame confidence κ_B is computed from the eigenvalue separation:

$$\kappa_B = \min\left(1, 3 \frac{\lambda_3 - \lambda_1}{\lambda_1 + \lambda_2 + \lambda_3}\right). \quad (4)$$

The rendering line-width of the frame triad axes is set to 3.0 when $\kappa_B > 0.5$ and 1.5 otherwise, providing a visual cue for confidence.

2.3 Mapping Confidence and Residual

The mapping from atomistic fragment to CG bead introduces a positional error $\delta = |\mathbf{r}_{\text{COM}} - \mathbf{r}_{\text{COG}}|$. The mapping confidence is

$$c_B = \max\left(0, 1 - \frac{\delta}{2}\right), \quad (5)$$

which scales shell opacity: $\alpha_{\text{shell}} = \alpha_0 (0.3 + 0.7 c_B)$. Poorly mapped beads become visually faint, drawing the researcher’s attention to reliable regions.

2.4 Correspondence with the Canonical Container Geometry

The Identity–State Decomposition Framework (§0) defines for every particle a *Canonical Container Geometry Approximation*:

$$\Omega_i = \left(R_i, \mathcal{T}_i(\theta, \phi), \rho_i(r)\right) \quad (\S 0, \text{Eq. ??})$$

Each component of Ω_i has a direct, implemented correspondence in the Fuzzy Ball Model.

Effective radius $R_i \longleftrightarrow R_{\text{eff}}$. The §0 definition, $R_i = \alpha_Z Z_i^{1/3} + \alpha_Q |Q_i|$, provides a *first-principles* estimate suitable for isolated atoms. In the CG context, the bead aggregates many atoms, so the Fuzzy Ball Model uses a *probe-radius* or mass-derived fallback (Equations ??–??). Both converge to the same role: determining the shell extent displayed by Layer 1. The §0 radius governs the formal interaction envelope; the visualisation radius governs what the researcher sees. They share the invariant:

$$R_{\text{eff}} \propto R_i \propto V_{\text{frag}}^{1/3}, \quad (6)$$

where V_{frag} is the volumetric extent of the mapped fragment.

Angular texture $\mathcal{T}_i(\theta, \phi) \longleftrightarrow S^{(k)}(\theta, \varphi)$. The §0 angular texture is a single spherical-harmonic expansion that encodes the directional interaction shape of the canonical container. The Fuzzy Ball Model implements a *three-channel* generalisation:

$$\mathcal{T}_i(\theta, \phi) \longrightarrow \{S^{(0)}, S^{(1)}, S^{(2)}\}(\theta, \varphi), \quad (7)$$

one expansion per physical channel (steric, electrostatic, dispersion). The synthesised channel $S^{(\text{syn})}$ (Equation ??) recovers the single-scalar §0 texture as a weighted contraction. The implementation uses the same mathematical object—real spherical harmonics Y_ℓ^m —at both levels, so the correspondence is exact up to the number of channels.

Radial density envelope $\rho_i(r) \longleftrightarrow$ **radial exaggeration** ξ . The §0 radial envelope, $\rho_i(r) = \rho_0 e^{-r/\lambda_i}$, describes a soft exponential decay of the interaction field. In the Fuzzy Ball Model, this manifests as the *radial exaggeration* parameter ξ (Equation ??), which deforms the rendered surface proportionally to the local field strength. When $\xi > 0$ the shell bulges outward in directions of high $S^{(k)}$, reproducing the intuition that $\rho_i(r)$ is larger in directions where the interaction field is strong. The formal mapping is:

$$\rho_i(R_{\text{eff}} \cdot \hat{\mathbf{n}}(\theta, \varphi)) \sim 1 + \xi \bar{S}^{(k)}(\theta, \varphi), \quad (8)$$

providing a visual analogue of the radial density profile.

Summary. The correspondence is collected in Table ??.

Table 2: Mapping between §0 Canonical Container Geometry Ω_i and the Fuzzy Ball Model implementation.

§0 Component	Fuzzy Ball Implementation	Source
R_i (effective radius)	<code>effective_radius</code>	<code>bead_visual_record.hpp</code>
$\mathcal{T}_i(\theta, \phi)$ (angular texture)	$S^{(k)}(\theta, \varphi)$ via SH expansion	<code>SurfaceStateGrid</code>
$\rho_i(r)$ (radial density)	Radial exaggeration ξ	<code>radial_exaggeration</code>
Σ_i (structural role)	<code>structural_role</code>	<code>StructuralRole</code> enum
Λ_i (stability class)	<code>stability_class</code>	<code>StabilityClass</code> enum

2.5 Structural Role and Stability Class

The Identity–State Decomposition Framework assigns each particle two discrete classification fields: a *structural role* Σ_i and a *stability class* Λ_i . Both are now carried on every `Bead` and propagated into the `BeadVisualRecord` for rendering-time inspection.

Structural role Σ_i . A five-valued prior encoding dominant bonding topology:

$$\Sigma_i \in \{0=\text{inert}, 1=\text{ionic}, 2=\text{directional covalent}, 3=\text{metallic}, 4=\text{mixed}\}.$$

The role biases the three-channel interaction kernel via per-channel weights $w_k^{(\Sigma)}$ (Table ??). For a bead pair (A, B) the combined weight uses a geometric mean, analogous to Lorentz–Berthelot mixing:

$$w_k^{AB} = \sqrt{w_k^A \cdot w_k^B}. \quad (9)$$

Table 3: Structural role Σ_i : per-channel weight biases. Values are multipliers on the existing coupling constants λ_k .

Σ	w_{steric}	w_{elec}	w_{disp}
0 — Inert	0.3	0.1	1.5
1 — Ionic-Dominant	0.5	1.5	0.3
2 — Directional Cov.	1.5	0.5	0.7
3 — Metallic	1.0	0.2	1.3
4 — Mixed	1.0	1.0	1.0

Stability class Λ_i . A four-valued persistence classification:

$$\Lambda_i \in \{0=\text{transient}, 1=\text{metastable}, 2=\text{ambient-stable}, 3=\text{bulk-lattice}\}.$$

Both fields are visible in the Comparison view mode and in the ImGui inspector panel, satisfying the anti-black-box requirement that every identity-state component of a bead be directly inspectable.

Implementation: `StructuralRole` and `StabilityClass` enums in `coarse_grain/core/bead.hpp`; the classifier `classify_structural_role()` infers Σ_i from the Z values of the parent atom group.

§3 Layer 2: Surface-State Descriptor Field

3.1 Spherical Harmonic Expansion

Each bead carries a *unified descriptor* consisting of three scalar fields on S^2 , one per interaction channel k :

$$S^{(k)}(\theta, \varphi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}^{(k)} Y_{\ell}^m(\theta, \varphi), \quad (10)$$

where $k \in \{0 = \text{steric}, 1 = \text{electrostatic}, 2 = \text{dispersion}\}$ and Y_{ℓ}^m are real spherical harmonics.

The coefficient vectors $\{c_{\ell m}^{(k)}\}$ are pre-solved during bead construction and stored in the `UnifiedDescriptor`. At visualisation time they are evaluated on a structured angular grid.

3.2 Surface-State Grid

The Layer 2 grid tessellates S^2 into $(N_\theta + 1) \times N_\phi$ vertices in row-major order. The default resolution is $N_\theta = 24$, $N_\phi = 48$, yielding 1 200 vertices:

$$\theta_i = \frac{\pi i}{N_\theta}, \quad i = 0 \dots N_\theta, \quad \varphi_j = \frac{2\pi j}{N_\phi}, \quad j = 0 \dots N_\phi - 1. \quad (11)$$

At each vertex, all three channels are evaluated via Equation ???. A fourth *synthesised* channel combines them:

$$S^{(\text{syn})}(\theta, \varphi) = w_s S^{(0)} + w_e S^{(1)} + w_d S^{(2)}, \quad (12)$$

with equal default weights $w_s = w_e = w_d = \frac{1}{3}$.

Per-channel value ranges $[S_{\min}^{(k)}, S_{\max}^{(k)}]$ are tracked for colourmap normalisation. The active channel is selectable at runtime.

3.3 Surface-State Channels

Table 4: Surface-state descriptor channels.

k	Name	Physical Meaning
0	Steric	Geometric accessibility / excluded-volume envelope
1	Electrostatic	Charge-field distribution from partial charges
2	Dispersion	Attractive interaction strength (London dispersion)
syn	Synthesized	Weighted combination of all three channels

3.4 Rendering the Heatmap

The grid is rendered as a quad-strip tessellation over the shell surface. Each vertex colour is obtained by mapping the active channel’s value through a sequential colourmap, normalised by the tracked min/max range.

Optional *radial exaggeration* ξ deforms the sphere in the normal direction:

$$\mathbf{r}(\theta, \varphi) = \mathbf{r}_B + R_{\text{eff}} \left(1 + \xi \bar{S}^{(k)}(\theta, \varphi) \right) \hat{\mathbf{n}}(\theta, \varphi), \quad (13)$$

where $\bar{S}^{(k)}$ is the normalised channel value in $[0, 1]$. When $\xi = 0$ the surface remains a perfect sphere; larger values produce a “bumpy” surface proportional to the field strength.

§4 Layer 3: Internal Structural Reference

Layer 3 renders the atomistic fragment that was mapped into the bead. This allows direct visual comparison between the CG representation and its atomistic source.

4.1 Source Fragment Miniature

The `SourceFragmentMiniature` stores a lightweight copy of the atomistic geometry:

- **Atoms:** position, atomic number, Van der Waals radius (simplified lookup), and topological flags.
- **Bonds:** atom index pair (i, j) and bond order.
- **Centres:** centre-of-mass $\mathbf{r}_{\text{COM}}$ and centre-of-geometry $\mathbf{r}_{\text{COG}}$.

4.2 Internal Reference Rendering Options

Table 5: Configurable options for Layer 3 rendering.

Option	Description	Default
<code>show_atoms</code>	Draw atom spheres	true
<code>show_bonds</code>	Draw bond sticks	true
<code>show_orientation_markers</code>	Draw local frame axes	true
<code>element_coloring</code>	CPK element colours	true
<code>atom_scale</code>	Fraction of VdW radius	0.25
<code>bond_width</code>	Bond line width (px)	1.5

§5 View Modes

The eight view modes control the composition of Layers 1–3 and additional diagnostic overlays.

Table 6: View-mode composition matrix. α denotes shell opacity. “Triad” = local frame triad. “Heatmap” = Layer 2 surface-state field. “Fragment” = Layer 3 atomistic reference. “Halo” = residual glyph.

Mode	α	Triad	Scaffold	Heatmap	Fragment	Halo
Shell	0.60					
Scaffold	0.25	✓	✓			
SurfaceState	0.10	✓		✓		
InternalRef	0.12	✓			✓	
Skeleton	0.15	✓			✓	
Cutaway	0.50	✓	✓			
Residual	0.30	✓				✓
Comparison	0.15	✓		✓	✓	✓

Shell. The simplest mode. A semi-transparent sphere at R_{eff} , colour-coded by chemical class: purple for aromatics, gold for metals, teal for cyclic fragments, blue for all others. Opacity encodes mapping confidence.

Scaffold. Adds the solved direction vectors (coordination geometry) as radial lines from the bead centre, plus the surface descriptor patch field rendered as directional glyphs.

SurfaceState. The Layer 2 heatmap mode. The shell becomes a ghost ($\alpha = 0.10$) and is overlaid with the spherical descriptor field. The active channel is selectable (steric, electrostatic, dispersion, or synthesised).

InternalRef. The Layer 3 mode. The atomistic fragment is rendered inside the ghost shell using ball-and-stick representation with CPK colouring.

Skeleton. Similar to InternalRef but displays the raw stick model of the source fragment.

Cutaway. A clipping plane slices the shell to expose internal structure. The scaffold and descriptor patches are drawn within the visible hemisphere.

Residual. Displays a residual halo whose size and colour encode the descriptor reconstruction residual, geometry distortion, and frame ambiguity. Used to identify beads where the CG mapping is least faithful.

Comparison. All diagnostic layers simultaneously: ghost shell, heatmap, atomistic fragment, frame triad, and residual halo. The most information-dense mode, intended for close-up single-bead analysis.

§6 Scale Hierarchy

Five scale levels govern the level of detail rendered for the bead population as a whole.

Table 7: Scale hierarchy levels.

Level	Name	Description
0	SceneCloud	Distant overview; beads as point sprites
1	Cluster	Intermediate; smooth shaded spheres
2	BeadDetail	Full shell rendering with opacity/colour
3	BeadInterior	Internal structure visible through ghost shell
4	FragmentCompare	Single-bead comparison with atomistic overlay

Transitions between levels are governed by camera distance and may be overridden for selected beads via the inspector panel.

§7 Alpha Polarisability Model Integration

The “fuzzy” character of the ball extends beyond visualisation into the physical model. The *alpha model* fits atomic polarisabilities using a parameterised form that includes *blob* terms:

$$\alpha_{\text{eff}} = f_{\text{blob}}(\alpha_0, Z, \text{block}), \quad (14)$$

where f_{blob} is controlled by three fitted parameters:

- `blob_f_lin` — linear fuzzy-shell mixing fraction,
- `blob_f_half` — half-shell contribution,
- `blob_f_full` — full-shell contribution.

These are determined by evolutionary stochastic optimisation against reference polarisability data (see `tools/fit_alpha_model.cpp`). Additional fitted parameters include:

Table 8: Alpha model parameters (Method D).

Parameter	Description
$k_{\text{period}}[1 \dots 7]$	Period-dependent scaling factors
$c_{\text{block}}[s, p, d, f]$	Orbital-block correction terms
c_{rel}	Relativistic correction coefficient
β_f	f -block contraction exponent
a_{f1}, a_{f2}, a_{f3}	Lanthanide/actinide amplitude terms
σ_{f1}, σ_{f2}	Gaussian widths for f -block
b_{bind}	Binding-energy correction
q_{drude}	Drude oscillator effective charge

The blob parameters connect the visual “fuzziness” of the shell to a quantitative physical property: the polarisability of the electron cloud determines how “soft” and “diffuse” each bead appears, both in the force field and in the rendering.

§8 Environment-Responsive Coupling

The fuzzy ball adapts to its thermodynamic environment. The `BeadVisualRecord` captures an environment-state snapshot at each frame:

Table 9: Environment observables stored per bead.

Symbol	Field	Meaning
η_B	<code>eta</code>	Slow internal state (relaxation variable)
ρ_B	<code>rho</code>	Local bead density
f_B	<code>target_f</code>	Target function value
$P_{2,B}$	<code>P2</code>	Nematic order parameter
C_B	<code>coordination_C</code>	Coordination number

These observables modulate the three-channel kernel described in the Seed & Bead specification (γ_{steric} , γ_{elec} , γ_{disp}), closing the loop between simulation dynamics and visual representation.

§9 Connectivity and Neighbour Interactions

Each bead record stores a list of `NeighbourLink` entries that describe pairwise interactions with adjacent beads:

- `neighbour_bead_id` — index of the neighbour bead,
- `distance` — centre-to-centre separation (Å),
- `interaction_strength` — magnitude of the pairwise potential,
- `interaction_type` — string label classifying the geometry (e.g. “face-to-face”, “edge-to-face”).

This information enables graph-based analyses and can be exported to the snapshot reporting system for automated figure generation.

§10 Implementation Summary

10.1 Data Flow

The conversion from simulation data to visual output follows a deterministic pipeline:

1. **Bead construction.** The atomistic fragment is mapped to a CG bead via the **6+9 stepper**, producing a `Bead` with position, mass, surface descriptor, and unified descriptor.
2. **Record building.** `build_visual_record()` converts the `Bead` into a `BeadVisualRecord`, computing effective radius, local frame, anisotropy, descriptor grids, and source-fragment miniature.
3. **Inspector rendering.** `BeadInspectorView` consumes the record and dispatches to the selected view mode, composing Layers 1–3 with the appropriate opacity and overlay settings.

10.2 Key Source Files

Table 10: Implementation files for the fuzzy ball model.

File	Role
<code>coarse_grain/vis/bead_visual_record.hpp</code>	Data model: <code>BeadVisualRecord</code> , enumerations, grid types, <code>build_visual_record()</code>
<code>coarse_grain/vis/bead_inspector_view.hpp</code>	Renderer: view-mode dispatch, shell/scaffold/heatmap/ fragment/residual compositing
<code>tools/fit_alpha_model.cpp</code>	Alpha polarisability fitting with blob parameters
<code>tools/auto_fit_alpha.cpp</code>	Evolutionary optimiser for <code>AlphaModelParams</code>
<code>coarse_grain/models/seed_bead_stepper.hpp</code>	6+9 step function that produces bead data consumed by the visual record builder

10.3 Descriptor Evaluation: Spherical Harmonics

The inline evaluator computes real spherical harmonics $Y_\ell^m(\theta, \varphi)$ up to $\ell_{\max}$ and contracts with the coefficient vector:

```
inline double evaluate_sh_expansion_at(
    const std::vector<double>& coeffs,
    int l_max,
    double theta, double phi)
{
    auto Y = evaluate_all_harmonics_dynamic(theta, phi, l_max);
    double val = 0.0;
    int n = std::min((int)coeffs.size(), (int)Y.size());
    for (int i = 0; i < n; ++i)
        val += coeffs[i] * Y[i];
    return val;
}
```

§11 Results and Discussion

The fuzzy ball model provides several concrete improvements over conventional CG visualisation:

1. **Directional information is preserved.** The three-channel surface-state heatmap reveals steric, electrostatic, and dispersion anisotropy that would be invisible with uniform spheres.
2. **Mapping quality is visually encoded.** Shell opacity and residual halos immediately flag beads whose CG representation deviates significantly from the atomistic source.
3. **Multi-scale inspection is seamless.** The five-level scale hierarchy and eight view modes allow a researcher to navigate from a macroscopic scene overview to single-fragment comparison without switching tools.
4. **Physical fuzziness is quantified.** The alpha-model blob parameters tie the visual softness of each shell to a fitted atomic polarisability, connecting rendering to a measurable physical property.
5. **Environment responsiveness is captured.** The environment-state snapshot $(\eta_B, \rho_B, P_{2,B}, C_B, f_B)$ modulates the descriptor kernel, ensuring the visual representation reflects local thermodynamic conditions.

§12 Conclusion

The Fuzzy Ball Model transforms the conventional point-sphere representation of coarse-grained beads into a scientifically rich, multi-layer visual instrument. By encoding directional descriptor fields via spherical harmonic expansion, exposing the internal atomistic skeleton through progressive transparency, and coupling shell appearance to environment state and fitted polarisability parameters, the model enables deterministic, inspectable, and research-grade visualisation.

The eight-mode composition system and five-level scale hierarchy provide a systematic framework for navigating between overview and detail, supporting both

qualitative exploration and quantitative analysis. All outputs are fully deterministic and exportable for inclusion in scientific reports.

Future work includes adaptive tessellation based on descriptor gradient magnitude, time-dependent surface-state animation for trajectory playback, and integration of the fuzzy ball rendering with the automated snapshot and graph reporting pipeline.